

```
In [312... import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import re
```

```
In [313... # data importing
```

```
In [314... data = pd.read_csv("C:\\Users\\anshu kumar\\Downloads\\messy_customer_sales_data
```

```
In [315... # data inspection
data
```

```
Out[315...      Customer_ID  Name  Gender  Age  City  Signup_Date  Last_Purchase_
```

0	CUST4371	Paul Wilson	m	52.0	KOLKATA	2025-06-26	2025-0
1	CUST5957	Jason Thomas	M	51.0 years	NaN	2021-02-17	2025-0
2	CUST3754	Brittney Martinez	F	62.0	hyderabad	2023-11-05	2024-
3	CUST2934	Brenda Pierce	FEMALE	40.0	hyderabad	2022-03-13	2025-
4	CUST5683	Matthew Carroll	f	41.0	CHENNAI	2024-04-05	2024-
...	...	...	...	...	...	...	...
10195	CUST10767	Robert Lewis	female	35.0 years	delhi	2020-12-08	2025-0
10196	NaN	Diane Evans	M	53.0	bangalore	2023-12-31	2025-0
10197	CUST6315	Joshua Martinez	m	25.0	hyderabad	2022-02-15	2025-0
10198	CUST4812	Sarah Miller	FEMALE	55.0	NaN	2021-03-16	2025-0
10199	CUST6588	David Potter	female	34.0	HYDERABAD	2020-10-12	2025-0

10200 rows × 12 columns



```
In [316... # data info
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10200 entries, 0 to 10199
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Customer_ID           9177 non-null   object
1   Name                  10200 non-null  object
2   Gender                9174 non-null   object
3   Age                  9249 non-null   object
4   City                  9184 non-null   object
5   Signup_Date           10200 non-null  object
6   Last_Purchase_Date     9188 non-null   object
7   Purchase_Amount       9179 non-null   float64
8   Feedback_Score        9177 non-null   float64
9   Email                 10200 non-null  object
10  Phone_Number          10200 non-null  int64
11  Country               9468 non-null   object
dtypes: float64(2), int64(1), object(9)
memory usage: 956.4+ KB
```

In [317...

```
# data describe
data.describe().round()
```

Out[317...

	Purchase_Amount	Feedback_Score	Phone_Number
count	9179.0	9177.0	1.020000e+04
mean	29090.0	5.0	4.979974e+09
std	208697.0	3.0	2.902593e+09
min	-500.0	1.0	9.208990e+05
25%	12295.0	3.0	2.449157e+09
50%	24330.0	5.0	4.988639e+09
75%	37130.0	8.0	7.510448e+09
max	999999.0	10.0	9.994402e+09

In [318...

```
# data null values
data.isnull().sum()
```

Out[318...

```
Customer_ID      1023
Name              0
Gender           1026
Age              951
City             1016
Signup_Date       0
Last_Purchase_Date 1012
Purchase_Amount   1021
Feedback_Score    1023
Email            0
Phone_Number      0
Country           732
dtype: int64
```

In [319...

```
# data duplicate rows shape
data[data.duplicated()].shape
```

Out[319... (15, 12)

```
In [320... # data duplicate rows
data[data.duplicated()]
```

Out[320... Customer_ID Name Gender Age City Signup_Date Last_Purchase_I

2032	CUST2755	Casey Campbell	male	NaN	Hyderabad	2023-12-01	2024-1
3191	NaN	Travis Schneider	f	51.0	NaN	2025-04-20	2025-0
4312	CUST8305	Stanley Cain	FEMALE	51.0	Bangalore	2023-08-14	2024-1
5611	CUST6288	Jonathon Kim	male	46.0	Chennai	2023-11-22	2025-0
5680	CUST2695	Lisa Durham	NaN	46.0	Chennai	2021-04-18	2024-10
5899	CUST8841	Amanda Hill	NaN	34.0	chennai	2022-09-11	2025-0
6070	CUST10780	Emily Smith	NaN	23.0	Kolkata	2023-12-09	2024-10
6240	CUST9745	David Morales	NaN	56.0	NaN	2023-02-18	2024-1
6425	CUST4631	George Villa	m	26.0	Mumbai	2022-05-21	2025-0
6498	CUST1711	Stephanie Elliott	NaN	44.0	MUMBAI	2023-05-17	2025-0
7118	CUST9380	Paul Wilson	MALE	42.0	Bangalore	2025-07-31	2025-0
8392	CUST1436	Ralph Anderson	female	33.0	Kolkata	2022-10-21	2025-0
9155	CUST10040	Mark Taylor	NaN	43.0	Delhi	2021-04-16	2025-0
9960	CUST6220	John Rodriguez	male	51.0	Hyderabad	2021-03-28	2025-0
10140	CUST4367	Amber Kennedy	F	61.0	Bangalore	2023-11-05	2025-0



```
In [321... data[data['Customer_ID'] == 'CUST2755']
```

Out[321...

	Customer_ID	Name	Gender	Age	City	Signup_Date	Last_Purchase_Da
1874	CUST2755	Casey Campbell	male	NaN	Hyderabad	2023-12-01	2024-11-01
2032	CUST2755	Casey Campbell	male	NaN	Hyderabad	2023-12-01	2024-11-01

In [322...

```
# duplicate column
data['Customer_ID'].value_counts().head(177)
```

Out[322...

```
Customer_ID
CUST3374      2
CUST7749      2
CUST10800     2
CUST6588      2
CUST9062      2
..
CUST7333      2
CUST4492      2
CUST2872      2
CUST3501      2
CUST10780     2
Name: count, Length: 177, dtype: int64
```

In [323...

```
data[data['Customer_ID'] == 'CUST3374']
```

Out[323...

	Customer_ID	Name	Gender	Age	City	Signup_Date	Last_Purchase_Date
2914	CUST3374	Timothy Wallace	m	58.0	Bangalore	2024-03-08	2025-08-21
3047	CUST3374	Timothy Wallace	m	58.0	bangalore	2024-03-08	2025-08-21

In [324...

```
data['Age'].value_counts()
```

Out[324...

```
Age
66.0      197
62.0      194
43.0      188
34.0      186
64.0      185
...
33.0 years    4
-10           3
39.0 years    3
32.0 years    3
56.0 years    2
Name: count, Length: 108, dtype: int64
```

In [325...

```
for col in data.columns:
    if data[col].nunique() < 20:
        print(data[col].value_counts())
        print('-'*50)
```

```
Gender
f      1184
M      1171
m      1163
F      1157
MALE   1131
female 1128
male   1121
FEMALE 1119
Name: count, dtype: int64
```

```
-----
City
Kolkata      820
Mumbai       812
Chennai      784
Bangalore    773
Hyderabad    770
Delhi        763
CHENNAI      404
KOLKATA      395
MUMBAI       393
hyderabad    384
bangalore    383
DELHI        378
delhi        369
BANGALORE    363
HYDERABAD    360
mumbai       352
chennai      343
kolkata      338
Name: count, dtype: int64
```

```
-----
Feedback_Score
2.0      952
4.0      947
7.0      938
6.0      927
3.0      913
8.0      912
1.0      907
9.0      903
10.0     901
5.0      877
Name: count, dtype: int64
```

```
-----
Country
India     7132
IND        793
india     772
InDia     771
Name: count, dtype: int64
```

Data Cleaning

In [326...

```
# remove null values
data.dropna(subset=['Customer_ID'], inplace = True)
```

In [327... `data.isnull().sum()`

```
Out[327... Customer_ID      0
Name            0
Gender          934
Age             859
City            918
Signup_Date     0
Last_Purchase_Date 914
Purchase_Amount 927
Feedback_Score  905
Email           0
Phone_Number    0
Country         664
dtype: int64
```

In [328... `data['Age'].unique()`

```
Out[328... array(['52.0', '51.0 years', '62.0', '40.0', '41.0', nan, '18.0',
      '43.0 years', '40.0 years', '26.0', '32.0', '22.0', '59.0', '65.0',
      '61.0', '31.0', '54.0 years', '55.0', '69.0', '61.0 years', '24.0',
      '63.0', '19.0', '50.0', '56.0', '36.0', '68.0', '43.0', '38.0',
      '27.0', '57.0 years', '23.0', '25.0', '66.0', '28.0', '30.0',
      '46.0', '48.0', '20.0', '37.0', '67.0', '51.0', '35.0', '58.0',
      '29.0', 'nan years', '39.0', '49.0', '47.0', '42.0', '44.0',
      '64.0', '53.0', '60.0', '59.0 years', '45.0', '21.0', '34.0',
      '54.0', '48.0 years', '46.0 years', '33.0', '57.0', '30.0 years',
      '58.0 years', '35.0 years', '34.0 years', '69.0 years', '250',
      '19.0 years', '27.0 years', '53.0 years', '65.0 years',
      '66.0 years', '44.0 years', '49.0 years', '25.0 years',
      '23.0 years', '62.0 years', '41.0 years', '33.0 years',
      '28.0 years', '22.0 years', '20.0 years', '42.0 years',
      '45.0 years', '3', '63.0 years', '37.0 years', '38.0 years',
      '55.0 years', '18.0 years', '36.0 years', '67.0 years',
      '29.0 years', '39.0 years', '31.0 years', '47.0 years',
      '52.0 years', '-10', '60.0 years', '24.0 years', '26.0 years',
      '21.0 years', '64.0 years', '50.0 years', '68.0 years',
      '32.0 years', '56.0 years'], dtype=object)
```

```
In [329... # clear object from age column
# 🚩 So basically:
# 🖐️ For every value in the Age column, it:
# Extracts only the numeric part
# Keeps original if no number exists
#Extracts only the numeric part
#Keeps original if no number exists

def extract_age(age):
    age_num = re.findall('[0-9]+', str(age))
    if len(age_num)>0:
        return age_num[0]
    else:
        return age

data['Age'] = data['Age'].apply(lambda x: extract_age(x))
```

In [330... `data_age = data[data['Age'] != 'nan years']['Age']`

```
In [331... age_median = int(data_age.dropna().astype('int64').median())
```

```
In [332... age_median
```

```
Out[332... 43
```

```
In [333... # replace 'nan years' to median
data.replace('nan years', age_median, inplace = True)
```

```
In [334... # fill null to median
data['Age'] = data['Age'].fillna(age_median)
```

```
In [335... # find unique in age column
data['Age'].unique()
```

```
Out[335... array(['52', '51', '62', '40', '41', 43, '18', '43', '26', '32', '22',
      '59', '65', '61', '31', '54', '55', '69', '24', '63', '19', '50',
      '56', '36', '68', '38', '27', '57', '23', '25', '66', '28', '30',
      '46', '48', '20', '37', '67', '35', '58', '29', '39', '49', '47',
      '42', '44', '64', '53', '60', '45', '21', '34', '33', '250', '3',
      '10'], dtype=object)
```

```
In [336... # fill null in purchase amount
data['Purchase_Amount'] = data['Purchase_Amount'].fillna(data['Purchase_Amount'])
```

```
In [337... data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 9177 entries, 0 to 10199
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Customer_ID            9177 non-null   object
1   Name                   9177 non-null   object
2   Gender                 8243 non-null   object
3   Age                   9177 non-null   object
4   City                   8259 non-null   object
5   Signup_Date            9177 non-null   object
6   Last_Purchase_Date     8263 non-null   object
7   Purchase_Amount        9177 non-null   float64
8   Feedback_Score         8272 non-null   float64
9   Email                  9177 non-null   object
10  Phone_Number            9177 non-null   int64
11  Country                 8513 non-null   object
dtypes: float64(2), int64(1), object(9)
memory usage: 932.0+ KB
```

```
In [338... # Signup_Date convert object to datetime
data['Signup_Date'] = pd.to_datetime(data['Signup_Date'])
```

```
In [339... data['Signup_Date'].dtype
```

```
Out[339... dtype('<M8[ns]')
```

```
In [340... # Last_Purchase_Date convert object to datetime
data['Last_Purchase_Date'] = pd.to_datetime(data['Last_Purchase_Date'])
```

```
In [341...] data['Last_Purchase_Date'].dtype
```

```
Out[341...] dtype('<M8[ns]')
```

```
In [342...] # convert age object to numerical
data['Age'] = pd.to_numeric(data['Age'], downcast = 'integer')
```

```
In [343...] data['Age'].dtype
```

```
Out[343...] dtype('int16')
```

```
In [380...] # fill multi column
for col in ['Gender', 'City', 'Country']:
    data[col].fillna(data[col].mode()[0], inplace = True)
```

C:\Users\anshu kumar\AppData\Local\Temp\ipykernel_19120\981913385.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.

The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or 'df[col] = df[col].method(value)' instead, to perform the operation inplace on the original object.

```
data[col].fillna(data[col].mode()[0], inplace = True)
```

```
In [381...] data['Feedback_Score'] = data['Feedback_Score'].fillna(data['Feedback_Score'].mode()[0])
```

```
In [383...] # forward fill by using ffill for Last_Purchase_Date
```

```
data['Last_Purchase_Date'].ffill(inplace = True)
```

```
In [347...] data.isnull().sum()
```

```
Out[347...] Customer_ID      0
            Name          0
            Gender        0
            Age           0
            City          0
            Signup_Date    0
            Last_Purchase_Date 0
            Purchase_Amount 0
            Feedback_Score 0
            Email         0
            Phone_Number   0
            Country        0
            dtype: int64
```

```
In [ ]:
```

fixing Inconsistent formatting

```
In [348... data['Gender'].value_counts()
```

```
Out[348... Gender
M          1990
m          1053
f          1051
F          1039
male       1017
MALE       1014
female     1010
FEMALE     1003
Name: count, dtype: int64
```

```
In [349... data['City'].value_counts()
```

```
Out[349... City
Kolkata      1657
Mumbai        721
Chennai       702
Delhi         697
Bangalore     692
Hyderabad     673
CHENNAI       370
hyderabad     352
bangalore     350
MUMBAI        349
DELHI         349
KOLKATA       343
HYDERABAD     340
delhi         330
BANGALORE     325
mumbai        316
chennai       308
kolkata       303
Name: count, dtype: int64
```

```
In [350... data['Country'].value_counts()
```

```
Out[350... Country
India       7090
IND          704
InDia       694
india       689
Name: count, dtype: int64
```

```
In [351... data['Gender'].unique()
```

```
Out[351... array(['m ', 'M', 'F', 'FEMALE', 'f ', 'male', 'MALE', 'female'],
      dtype=object)
```

```
In [352... # convert age in lower case
data['Gender'] = data['Gender'].str.lower().str.strip()
```

```
In [353... data['Gender'] = data['Gender'].replace({'m': 'male', 'f': 'female'})
```

```
In [354... data['Gender'].unique()
```

```
Out[354... array(['male', 'female'], dtype=object)
```

```
In [355... data['City'].unique()

Out[355... array([' KOLKATA ', ' Kolkata ', ' hyderabad ', ' CHENNAI ', ' kolkata ',
      ' BANGALORE ', ' Hyderabad ', ' HYDERABAD ', ' Mumbai ',
      ' chennai ', ' Delhi ', ' bangalore ', ' Bangalore ', ' delhi ',
      ' Chennai ', ' MUMBAI ', ' DELHI ', ' mumbai '], dtype=object)
```

```
In [356... # convert age in upper case
data['City'] = data['City'].str.upper().str.strip()
```

```
In [357... data['City'].unique()

Out[357... array(['KOLKATA', 'HYDERABAD', 'CHENNAI', 'BANGALORE', 'MUMBAI', 'DELHI'],
      dtype=object)
```

```
In [358... data['Country'].unique()

Out[358... array(['India', 'india', 'InDia', 'IND'], dtype=object)
```

```
In [359... # convert country in upper case
data['Country'] = data['Country'].str.upper()
```

```
In [360... data['Country'].unique()

Out[360... array(['INDIA', 'IND'], dtype=object)
```

```
In [361... # Replace IND to INDIA
data['Country'] = data['Country'].replace({'IND': 'INDIA'})
```

```
In [362... data['Country'].unique()

Out[362... array(['INDIA'], dtype=object)
```

```
In [ ]:
```

```
In [ ]:
```

Handling Duplicates

```
In [363... data[data.duplicated()].shape
```

```
Out[363... (163, 12)
```

```
In [364... # drop duplicates
data.drop_duplicates(inplace = True)
```

```
In [365... data[data.duplicated()].shape
```

```
Out[365... (0, 12)
```

```
In [366... data['Customer_ID'].value_counts()
```

```
Out[366... Customer_ID
CUST9534    2
CUST4196    2
CUST5341    2
CUST2510    2
CUST1002    2
..
CUST7618    1
CUST2436    1
CUST5982    1
CUST5432    1
CUST4812    1
Name: count, Length: 9000, dtype: int64
```

```
In [367... data[data['Customer_ID'] == 'CUST9534']
```

```
Out[367...      Customer_ID  Name  Gender  Age  City  Signup_Date  Last_Purchase_Date
3528  CUST9534  Melissa Jones  male  53  KOLKATA  2021-08-26  2025-08-11
9005  CUST9534  Melissa Jones  male  53  KOLKATA  2021-08-26  2025-01-05
```



```
In [368... data.drop_duplicates(subset = ['Customer_ID'], keep = 'first', inplace = True)
```

```
In [369... data['Customer_ID'].value_counts()
```

```
Out[369... Customer_ID
CUST4371    1
CUST8487    1
CUST7407    1
CUST8555    1
CUST6567    1
..
CUST9023    1
CUST5769    1
CUST6395    1
CUST4323    1
CUST4812    1
Name: count, Length: 9000, dtype: int64
```

```
In [370... data.shape
```

```
Out[370... (9000, 12)
```

```
In [ ]:
```

Handling outlier

```
In [ ]:
```

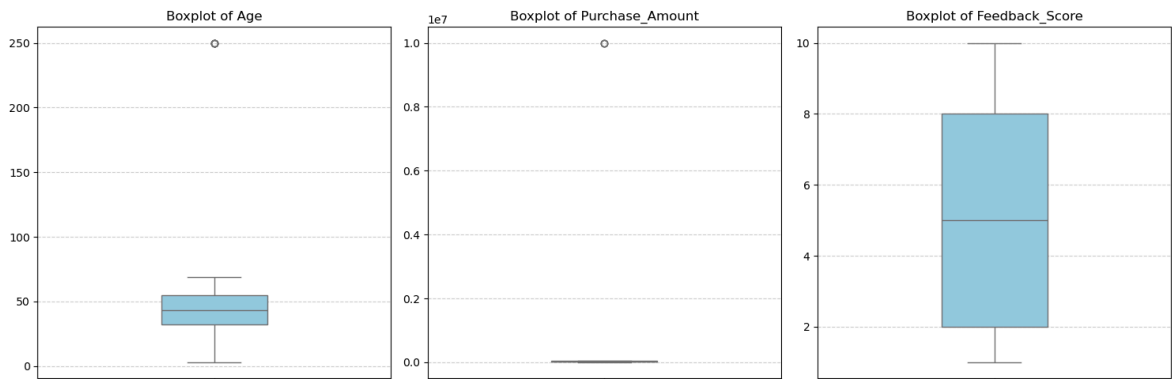
```
In [378... cols = ['Age', 'Purchase_Amount', 'Feedback_Score']
plt.figure(figsize=(15,5))
```

```

for i, col in enumerate(cols, 1):
    plt.subplot(1,3,i)
    sns.boxplot(y=data[col], color='skyblue', width=0.3)
    plt.title(f'Boxplot of {col}', fontsize=12)
    plt.ylabel('')
    plt.grid(axis='y', linestyle='--', alpha=0.6)

plt.tight_layout()
plt.show()

```



```
In [384... from scipy import stats
```

```
In [385... z_scores = np.abs(stats.zscore(data[['Age', 'Purchase_Amount']]))
```

```
In [386... data_clean = data[~(z_scores>3).any(axis = 1)]
```

```
In [387... data_clean
```

Out[387...

	Customer_ID	Name	Gender	Age	City	Signup_Date	Last_Purchase
0	CUST4371	Paul Wilson	male	52	KOLKATA	2025-06-26	2025-
1	CUST5957	Jason Thomas	male	51	KOLKATA	2021-02-17	2025-
2	CUST3754	Brittney Martinez	female	62	HYDERABAD	2023-11-05	2024-
3	CUST2934	Brenda Pierce	female	40	HYDERABAD	2022-03-13	2025-
4	CUST5683	Matthew Carroll	female	41	CHENNAI	2024-04-05	2024-
...	...	...	...	...	...	...	...
10193	CUST6352	Isaiah Terry	female	26	KOLKATA	2023-03-04	2025-
10194	CUST6146	Cody Thompson	female	40	KOLKATA	2024-08-21	2024-
10195	CUST10767	Robert Lewis	female	35	DELHI	2020-12-08	2025-
10197	CUST6315	Joshua Martinez	male	25	HYDERABAD	2022-02-15	2025-
10198	CUST4812	Sarah Miller	female	55	KOLKATA	2021-03-16	2025-

8989 rows × 12 columns



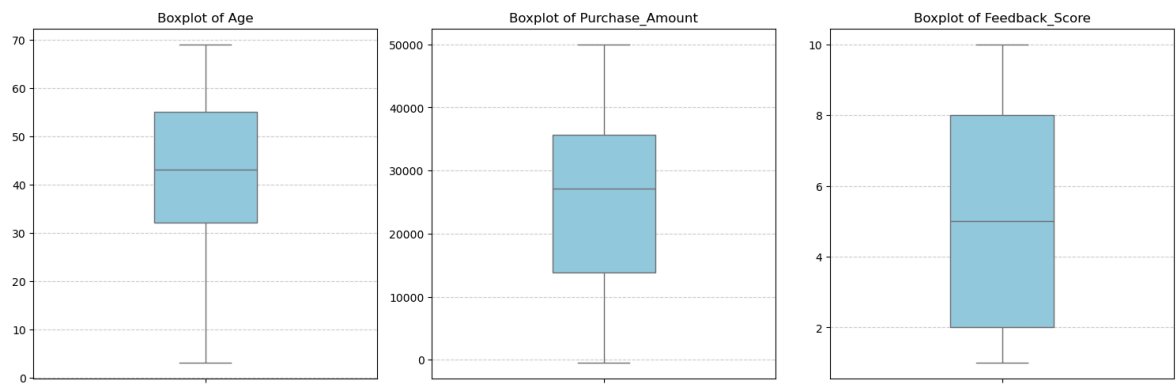
In [388...

```
cols = ['Age', 'Purchase_Amount', 'Feedback_Score']

plt.figure(figsize=(15,5))

for i, col in enumerate(cols, 1):
    plt.subplot(1,3,i)
    sns.boxplot(y=data_clean[col], color='skyblue', width=0.3)
    plt.title(f'Boxplot of {col}', fontsize=12)
    plt.ylabel('')
    plt.grid(axis='y', linestyle='--', alpha=0.6)

plt.tight_layout()
plt.show()
```



```
In [389... data_clean.to_csv('messy_clean_data', index= False)
```

```
In [ ]:
```